LEF MODEL · CHAPTER 20

Chapter 20: The Steam Ring of Mars, the Drainage through Valles Marineris and the MMX Mission Test

Based on Paper 10 — Mars: The Protrusion and the Ring (Parts F–O and Appendix B: the dynamics of the ring, Valles Marineris, the grooves of Phobos and the predictions for 2027–2031)

01 · THE MYSTERY

The mystery: how did one ring shape two opposite moons — and what proof will the MMX mission bring in 2031?

In Chapter 19 we saw how Pavonis Mons, standing at the top of the pointed bulge of Tharsis (4,450 km above the equator), released **Deimos** into a long elliptical orbit (10,170–23,990 km, e = 0.41, inclination ~ 10°) at 7.3 hours, and **Phobos** into a circular orbit (e = 0.002) on the corotation line (9,870 km) at 8.3 hours. Three critical closing mysteries now remain:

1. **Opposite fates in the same ring (F.1–F.6, G):** why did Deimos become circular, align with the equator, wear down to a diameter of 12.4 km and recede to 23,463 km, while the heavier and faster Phobos did not escape outward with the tide, kept its grooves, and is falling today from 19,100 km to 9,375 km?
2. **The enormous scar of Valles Marineris and the water balance (H.1–H.6, M):** what is the role of the Noctis Labyrinthus maze, the Valles Marineris canyon

and the outflow channels to Chryse — and where is the water of young Mars hiding?

3. **The laboratory decision of the MMX mission (Part III and Appendix B):** the **MMX** mission of the Japanese space agency JAXA (launch in October 2026, landing on Phobos in 2027 and return of samples to the Earth in **2031**) will measure the isotopes, the centre of mass and the grooves of Phobos and decide unambiguously between the LEF model and the impact and capture theories.

02 · AN IMAGE

An image: sandpaper on a crossing orbit, a headwind leash, and the balls in a ball bearing

Four tangible images illuminate the dynamics of the steam ring, the drainage scar and the marks of the vent on Phobos:

- **A body crossing a circular stream like sandpaper, against a boat sailing with the current (F.2–F.3):** before the ejections, Mars built **a round equatorial ring of steam and dust** that condensed into ice grains beyond the corotation line (and coalesced into clumps beyond the Roche limit for ice at 13,000 km). Deimos, thrown into a long elliptical orbit (e = 0.41), crossed the ring on every orbit at a relative speed of hundreds of metres per second: the ring **wore it down from 20.7 to 12.4 km** (a loss of 76%–83% of its volume), flattened its inclination from 10° to 1.8°, and sped it up at the far end (where it moved slower than the ring) until it circularised it and moved it out to 14,300 km (and from there the tide of the soft Mars, 8–20 times stronger than today, carried it to 23,463 km within 150–370 million years). Phobos, by contrast, entered a circular orbit at once at the speed of the ring itself — so the ring could neither speed it up nor wear it down!
- **"The gas leash" — a faint headwind balanced against the tidal push (F.4–F.6):** at the inner edge of the ring, gas held partly by its own pressure moves a little slower than orbital speed, creating **a headwind of ~ 100 m/s**. At a tiny gas density of **3–8 × 10^{-8} kg/m³** (as thin as the Earth's air at a height of 115 km), the wind's drag exactly balances the tiny tidal acceleration (~ 10^{-11}

m/s²). So the inner edge of the ring served as **a "leash"** that held Phobos close to the corotation line and led it outward slowly together with the line as Mars slowed — until the steam stopped and the ring dried out from the outside inward **3.5–3.9 billion years ago** (when Phobos stood at ~ 19,100 km and the Martian day had lengthened to ~ 22 hours)!

- **Trapped rocks like the balls in a ball bearing, and the knobs of a volcano's vent (K.1–K.3):** Phobos was the second, stiffer cork, pushed hard under pressure through the narrow vent of Pavonis (20–23 km wide). Stiff rock knobs in the vent wall (like the solid spines extruded from the vent of Mount St. Helens in 2004–2006) carved into Phobos **the parallel grooves** along its direction of motion (which curved slightly as the hot body rounded in space); and rocks trapped in the narrow gap between the cork and the wall rolled like **the balls in a ball bearing** and pressed into the soft surface **the chains of pits**, evenly spaced and always parallel to the grooves without ever crossing them!
- **The drainage system of a sinking basin (H.2–H.4):** when the bulge sank and lay down westward, the liquid and viscous contents inside it flowed east along the axis of the bulge: from the broken "neck" of **Noctis Labyrinthus**, through the trough of **Valles Marineris**, which descends 10 km from west to east (from +4.9 km to −5.2 km), filled an inner lake at a uniform level of −3,560 m, and spilled east through the Eos Chasma sill and the Simud and Tiu outflow channels onto the Chryse plain!

03 · SCIENCE AND MODEL

Science and model: the evolution of Phobos, the Pavonis caldera and the surface records (Parts F–M)

Parts F–M of Paper 10 connect every observational detail on Mars and its two moons into one physical sequence:

1. **Reconstructing the fall of Phobos and the origin of the ratio of 3.2 orbits per day (G, K.7–K.8):** the moment the ring dried out 3.9 Ga ago, when Phobos stood at 19,100 km (an orbital period of 22.3 hours against a Martian day of ~ 22 hours — that is, **exactly one orbit per day**), the corotation line

passed beyond it and Phobos began to fall inward. Since the tidal force weakens with the 5.5th power of distance, in the first 3 billion years it dropped only 3,800 km (about 4 cm per century), and in the last 400 million years it plunged more than 4,000 km (up to 1.8 m per century today at 9,375 km, orbital period 7.65 hours — a speed-up of 2.9 times, which together with the slowing of Mars from 22 to 24.6 h by 1.1 times gives exactly the ratio 2.9 × 1.1 = 3.2 orbits per day!).

2. **The volume balance of the Pavonis caldera and the record of Stickney (I.1–I.3, K.4–K.6):** the caldera at the top of Pavonis Mons is a single cylindrical pit, round and steep, 47 km wide and almost 5 km deep (depth/width ratio 0.10, the deepest of the four Tharsis volcanoes), with a volume of **8,300–8,700 km³** — matching the total volume of the two corks ejected from it (~ 6,740 km³ today) after the collapse of the roof of the shallow chamber. At the blunt end of Phobos lies the hollow of **Stickney** (9 km), formed when the leading blunt end struck the material in the mouth of the vent: a measurement on the shape model of Ernst et al. (2023) at 36 m resolution proves that the grooves on the floor of Stickney are **squashed and wider** (sharpness 0.34 against 0.44 outside it — lower than 89% of 80 random sites on Phobos) with no regolith cover, that is, the grooves were carved in the vent *before* Stickney was pressed in on exit!

3. **Solving the double water balance of Mars and the source of the global dust (H.5–H.6, M):** the water ejected through the corotation line into the ring was broken up by solar radiation and left a 6–7-fold enrichment of deuterium in the atmosphere (a global layer of at least 137 m, Villanueva et al. 2015), and the ring's ground-up dust settled as the uniform global dust of Mars; while most of the inner water of young Mars **never left** and stayed sealed beneath the same impermeable crust — in the pores of the middle crust at a depth of **11.5–20 km** discovered by the InSight lander (Wright et al. 2024, equivalent to a global layer 1–2 km deep)!



The diameter of the two moons (km) at ejection and today: Deimos, which crossed the heart of the ring, lost about 40% of its diameter; Phobos, which floated at the gaseous edge, hardly changed.

04 · THE MECHANISM IN WORDS

The mechanism in words: how the ring held Phobos on the corotation line and reversed the two inclinations

Parts F.1–F.6 and G of Paper 10 show how one physical interaction — an equatorial ring of steam and dust fed by the water ejected from Mars — explains both the extreme differences between Phobos and Deimos and the precise timing of Phobos' fall:

1. **The balance of forces in the inner "gas leash" (§F.4):** at the inner edge of the ring, close to the corotation line, the gas is held partly by its own pressure and so rotates a little slower than the Keplerian orbital speed. So Phobos meets **a faint headwind at a relative speed v ≈ 100 m/s**. At a gas density of $\rho_g = 3\text{–}8 \times 10^{-8}$ **kg/m³** (as thin as the Earth's atmosphere at a height of 115 km), the gas drag exactly balances the tidal acceleration pushing Phobos outward (~ 10^{-11} m/s²). As long as Mars ejected steam and slowed, the corotation line moved outward and led Phobos with it up to ~ 19,100 km.

2. **The ring drying out from the outside inward and the release of the leash (§F.5–F.6):** the ring was fed only from its inner base on the corotation line (while beyond 13,000 km, the Roche limit for ice, the material coalesced into clumps that no longer functioned as a ring). When the crust of young Mars stiffened, differentiated and cooled, the flux of steam stopped and the ring

dried out from the outside inward **3.5–3.9 billion years ago** (a Martian day of ~ 22 hours). At that moment the corotation line passed beyond Phobos (19,100 km), and Phobos began to fall inward to its orbit today (9,375 km). Had the ring dried out early, when it was at 9,900 km, it would have fallen and broken up in less than 100 million years!

3. **The differences in wear and in crater depth (§J.1–J.2, §L):** Deimos crossed the heart of the ring and the clumps beyond 13,000 km for millions of orbits, so it lost 76%–83% of its volume (worn down from a diameter of ~ 20.7 km to 12.4 km), lost its original vent grooves, and its craters are three times shallower than those of Phobos (median depth 1.9% of the width against 5.5% on Phobos). Phobos, which floated gently at the inner gaseous edge, lost only about 1% (a layer ~ 40 m thick) and kept its grooves and chains of pits.

05 · THE KEY EQUATIONS

Academic explanation and the key equations

The balance between the drag of the headwind at the inner edge of the ring and the tidal push of young Mars, and the rate of tidal change of the orbital semi-major axis da/dt, are formulated as follows (§G, Appendix A):

$$F_{\text{drag}} = \frac{1}{2}\rho_g A v_{\text{rel}}^2 = F_{\text{tide}} \quad (\rho_g \approx 3\text{–}8\times10^{-8}\ \text{kg/m}^3,\ v_{\text{rel}} \approx 100\ \text{m/s})$$

$$\frac{da}{dt} = \pm 3\left(\frac{k_2}{Q}\right)\left(\frac{m}{M}\right)\left(\frac{R}{a}\right)^5 na$$

Where the tidal dissipation coefficient today is $k_2/Q = 1.816\times10^{-3}$ (Lainey et al. 2021) and on the young, hot and soft Mars it was 8–20 times larger, Deimos climbed from 14,300 km to beyond the final corotation line (20,428 km) within 150–370 million years, while Phobos plunged from 19,119 km 3.9 Ga ago to 9,375 km today, at a rate accelerating from 4 cm/century to 1.8 m/century.

06 · THE SIGNATURE

The signature in reality: the matrix of firm predictions for the MMX mission (2027–2031)

In Part III and Appendix B, Paper 10 sets out **16 quantitative falsification tests (B1–B16)** for upcoming space missions — first among them JAXA's **MMX** mission (launch in October 2026, landing on Phobos in 2027 and return of ≥ 10 grams of regolith to the Earth in 2031):

- **The decisive isotopic and chemical test (N.1, B1–B4):** in the Phobos samples that MMX will return, the isotopes of **oxygen (O), chromium (Cr) and titanium (Ti)** will be found **on the pure Mars line only**, with no component of a foreign impactor or asteroid at all; the volatile isotopes of **zinc (Zn) and potassium (K)** will show no evaporation shift (unlike a molten giant impact); and the grains will show the texture of **a quenched volcanic foam with closed pores and hydrated minerals** (and not glass spherules condensed from vapour).
- **The structure and gravity test of Phobos and Deimos (N.2–N.6, B5–B8):** the gravity field of Phobos will reveal **an offset of the centre of mass by 300–500 m toward Mars (the blunt end)**; reconstructing the shape of Phobos into an elongated body will straighten the families of parallel grooves; and MMX flybys near Deimos will confirm a Martian composition identical to Phobos.

The lab box: comparing Deimos and Phobos in the ring, the tidal timeline and the tests of the MMX mission (Paper 10, Parts F–O and Appendix B)

Property	Deimos (the first ejection)	Phobos (the second ejection)
Time of ejection and initial orbit	**Day of 7.3 h; a long elliptical orbit (10,170–23,990 km, e = 0.41, inclination ~ 10°)**	**Day of 8.3 h; directly into a circle on the corotation line (9,870 km, e = 0.002)**
Zone of action in the ring	Crossed **the whole heart of the ring** and the clumps beyond 13,000 km at hundreds of m/s	Floated gently at **the inner gaseous edge** ($\rho_g = 3–8 \times 10^{-8}$ kg/m³)

Wear and change in size	**Worn down from a diameter of ~ 20.7 km (6.8 × 10^{15} kg) to 12.4 km (lost 76%–83% of its volume)**	**Lost only ~ 1% (~ 40 m); remained 22.3 km in size (1.07 × 10^{16} kg)**
Density and surface	**1,471 kg/m³** (spongy upper cork); smooth and rounded, without grooves (median hollow depth 1.9%)	**1,853–1,887 kg/m³** (compact second cork); parallel grooves, chains of pits and Stickney (9 km)
Orbit and fate today	**23,463 km (e = 0.0003, i ≈ 1.8°, orbit 30.3 h) — beyond the line, receding**	**9,375 km (e = 0.0151, i = 1.09°, orbit 7.65 h) — inside the line, falling (1.8 m/century)**

Test code	Measurement and mission	Prediction of the LEF model (the bulge and the ring)	Prediction of the mainstream explanation (impact / capture)
B1–B3	**Isotopes of O, Cr, Ti, D/H and Zn, K in the MMX samples (2031)**	**The pure Mars line only, zero foreign component, and no evaporation shift in Zn, K**	**Impact: a mixture of Mars + a foreign body (30%–65%) and depletion of light isotopes; capture: an asteroid**
B4	**The grain texture of Phobos (MMX, 2031)**	**A quenched volcanic foam with closed pores and hydrated minerals**	**Glass spherules melted and condensed from vapour**
B5	The gravity field of Phobos (MMX, 2027)	**The centre of mass offset by 300–500 m toward Mars (the blunt end)**	No prediction of a centre-of-mass offset
B6–B7	Imaging of the grooves and pit chains (MMX, 2027)	Concentrated on the western side, straightening in an elongated body, **squashed on the floor of Stickney**; parallel chains that do not cross grooves	Tidal stresses or Stickney ejecta (no difference inside Stickney, or random crossing)
B10 / B14	Gravity beneath Pavonis / seismology beneath Tharsis	**A mass deficit** beneath the Pavonis caldera; a middle crust saturated with liquid water beneath all of Tharsis	An ordinary shield caldera; no link between the moons and the water trapped at depth

07 · NEXT

The door ahead

In Chapters 19–20 we extended the LEF principle beyond the Earth–Moon system and showed how the same physical mechanism works on **Mars**: a young, expanded and fast planet grew on its equator **the pointed bulge of Tharsis**, carried Pavonis Mons to a height of 4,450 km into zero-net-weight conditions, and

with routine Vulcanian pressure released **Deimos** (7.3 h) and **Phobos** (8.3 h) into **the condensed steam ring** — and the MMX mission, which will return samples from Phobos in 2031, will provide the decisive laboratory test!

Chapter 21

The Law of Moons in the Solar System — the Ruler of Birth Modes, the Pump Principle and the Atmospheric Lid →

← Chapter 19 · Prologue

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